

Practice Problems on Survival Analysis

Advanced Statistical Methods II

Bachelor's Third Year Statistics

AY 2026–27

Notation

Let T denote the event time, C the censoring time, $Y = \min(T, C)$ the observed time, and $\Delta = \mathbb{I}(T \leq C)$ the event indicator. The survival function is $S(t) = P(T > t)$, the cumulative hazard is $H(t) = \int_0^t h(u) du$, and for continuous survival times $S(t) = \exp\{-H(t)\}$.

Problems

Problem 1: Survival, cumulative hazard, density, and median

Suppose that the hazard function of a failure time T is

$$h(t) = \begin{cases} 0.05, & 0 < t \leq 6, \\ 0.10, & 6 < t \leq 12, \\ 0.02, & t > 12. \end{cases}$$

1. Find the cumulative hazard $H(t)$.
2. Find the survival function $S(t)$.
3. Find $P(T > 12)$.
4. Find the median survival time.
5. Write the density $f(t)$ in terms of $h(t)$ and $S(t)$.

Problem 2: Risk sets and event times

The following data contain survival times in months. Here $\Delta = 1$ denotes an observed event and $\Delta = 0$ denotes a right-censored observation.

Subject	1	2	3	4	5	6	7	8	9	10	11	12
Y_i	2	3	4	5	6	7	8	10	11	12	15	16
Δ_i	1	0	1	1	0	1	0	1	0	1	1	0

1. List the distinct event times.

2. For each event time t_j , compute the risk set size n_j and the number of events d_j .
3. Explain why censored observations at earlier times are not included in later risk sets.
4. Compute the Kaplan–Meier estimate $\hat{S}(t)$ at each event time.
5. Estimate the median survival time.

Problem 3: Likelihood under right censoring

Let $T_1, \dots, T_n$ be independent continuous event times with density $f(t; \theta)$ and survival function $S(t; \theta)$. We observe (Y_i, Δ_i) , where $Y_i = \min(T_i, C_i)$ and $\Delta_i = \mathbb{I}(T_i \leq C_i)$.

1. Derive the likelihood contribution of an individual with an exact observed event time.
2. Derive the likelihood contribution of an individual whose event time is right-censored at y_i .
3. Show that the full likelihood can be written as

$$L(\theta) = \prod_{i=1}^n f(y_i; \theta)^{\Delta_i} S(y_i; \theta)^{1-\Delta_i}.$$

4. Rewrite the likelihood in terms of $h(t; \theta)$ and $H(t; \theta)$.
5. Briefly explain why right censoring is not the same as missing completely at random.

Problem 4: Exponential model with censoring

Suppose $T \sim \text{Exponential}(\theta)$ with mean parameter $\theta > 0$, so that

$$f(t; \theta) = \theta^{-1} e^{-t/\theta}, \quad S(t; \theta) = e^{-t/\theta}, \quad t > 0.$$

Consider the data

Y_i	4	6	9	13	15	18	20	22
Δ_i	1	0	1	1	0	1	0	1

1. Write the censored likelihood.
2. Derive the MLE of θ .
3. Compute the numerical value of the MLE.
4. Interpret the role of censored observations in the MLE.

Problem 5: Weibull model under right censoring

Let T follow a Weibull distribution with shape $\alpha > 0$ and scale $\lambda > 0$:

$$S(t; \alpha, \lambda) = \exp \left[- \left(\frac{t}{\lambda} \right)^\alpha \right], \quad f(t; \alpha, \lambda) = \frac{\alpha}{\lambda} \left(\frac{t}{\lambda} \right)^{\alpha-1} \exp \left[- \left(\frac{t}{\lambda} \right)^\alpha \right].$$

Use the same data as in Problem 4.

1. Write the censored log-likelihood $\ell(\alpha, \lambda)$.
2. Derive the score equation for λ when α is fixed.
3. Show that the Weibull model reduces to the exponential mean model when $\alpha = 1$ and $\lambda = \theta$.
4. Write R code to maximize the log-likelihood numerically using `optim`.

Problem 6: Kaplan–Meier estimator and Greenwood’s formula

Use the data in Problem 2.

1. Construct the Kaplan–Meier table with columns $t_j, n_j, d_j, \hat{S}(t_j)$.
2. Compute Greenwood’s estimated variance

$$\widehat{\text{Var}}\{\hat{S}(t)\} = \hat{S}(t)^2 \sum_{t_j \leq t} \frac{d_j}{n_j(n_j - d_j)}.$$

3. Give an approximate pointwise 95% confidence interval for $S(10)$ using the plain normal approximation.
4. Explain one limitation of the plain normal interval for survival probabilities.

Problem 7: Kaplan–Meier as a diagnostic for parametric fit

For the data in Problem 4, suppose one wants to check whether an exponential model is plausible.

1. Fit the exponential model using the MLE from Problem 4 and write the fitted survival curve.
2. Describe how you would compare the fitted exponential curve with the Kaplan–Meier estimate.
3. Show why, under an exponential model, $-\log S(t)$ should be approximately linear in t through the origin.
4. Show why, under a Weibull model, $\log\{-\log S(t)\}$ should be approximately linear in $\log t$.
5. Write R code to make these two diagnostic plots using Kaplan–Meier estimates.

Problem 8: Cox partial likelihood by hand

Consider six subjects with one binary covariate Z .

Subject	A	B	C	D	E	F
Y_i	2	3	5	6	7	8
Δ_i	1	1	0	1	1	0
Z_i	1	0	1	1	0	0

Assume the Cox proportional hazards model

$$h(t | Z) = h_0(t) \exp(\beta Z).$$

There are no tied event times.

1. List the risk set at each event time.
2. There is a notion in Survival Analysis called the *Cox partial likelihood*. Write the Cox partial likelihood $L_p(\beta)$ in this problem.
3. Write the log-partial likelihood $\ell_p(\beta)$.
4. Derive the score equation.
5. Numerically, the solution is approximately $\hat{\beta} = 1.073$. Interpret the hazard ratio.

Problem 9: Hazard ratios and survival curves under proportional hazards

Suppose the Cox model with a single continuous covariate X is

$$h(t | X) = h_0(t) \exp(0.35X).$$

1. Interpret the hazard ratio for a one-unit increase in X .
2. Interpret the hazard ratio for a five-unit increase in X .
3. If $S_0(t) = \exp\{-H_0(t)\}$ denotes the baseline survival curve, show that

$$S(t | X) = S_0(t)^{\exp(0.35X)}.$$

4. If $S_0(24) = 0.70$, compute $S(24 | X = 2)$.
5. Explain why the proportional hazards assumption is about hazards, not survival probabilities.

Problem 10: R exercise - Kaplan–Meier, log-rank test, and Cox model

This exercise uses the `lung` dataset from the R package `survival`; it is in the spirit of applied R exercises such as those in [Moore \(2016\)](#).

1. Load the data and recode the status variable so that 1 denotes death and 0 denotes censoring.
2. Fit a Kaplan–Meier curve stratified by sex.
3. Estimate the median survival time in each group.
4. Perform a log-rank test comparing the two groups.
5. Fit a Cox model using sex, age, and ECOG performance score as covariates.
6. Interpret the coefficient of sex and the coefficient of age.

Problem 11: R exercise - simulation under a Cox-Weibull model

Generate data from a proportional hazards model with a Weibull baseline and one binary covariate.

1. Use the following code skeleton to simulate the data:

```
set.seed(202627)
n <- 250
z <- rbinom(n, size = 1, prob = 0.5)
beta <- log(1.8)
alpha <- 1.4
lambda <- 12
u <- runif(n)
Ttrue <- lambda * (-log(u) / exp(beta * z))^(1 / alpha)
C <- rexp(n, rate = 0.035)
time <- pmin(Ttrue, C)
status <- as.integer(Ttrue <= C)
dat <- data.frame(time = time, status = status, z = z)
```

2. Fit a Kaplan–Meier curve by z .
3. Fit a Cox model with z as the covariate.
4. Compare the estimated hazard ratio with the true hazard ratio.
5. Create a plot of $\log\{-\log \hat{S}(t)\}$ versus $\log t$ for the two groups and explain what approximately parallel curves would suggest.

Problem 12: A conceptual question on informative censoring

In a clinical study, patients are followed until relapse. Some patients are censored because the study ends. Others are censored because their health deteriorates and they stop attending follow-up visits.

1. Which censoring mechanism is more likely to be independent/non-informative?
2. Why can informative censoring bias the Kaplan–Meier estimator?
3. Give one practical strategy for detecting or reducing the impact of informative censoring.
4. Explain the connection with missing-data mechanisms such as MCAR, MAR, and MNAR.

References

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- Moore, D. F. (2016). *Applied Survival Analysis Using R*. Springer.

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Kalbfleisch, J. D. and Prentice, R. L. (2002). *The Statistical Analysis of Failure Time Data*. 2nd edition, Wiley.